Emotional Interaction

Interaction Design: Beyond Human-Computer Interaction

Detailed Study Notes

Based on Chapter 6 Presentation

Contents

1 Introduction to Emotional Interaction

Traditionally, Human-Computer Interaction (HCI) focused on **usability**: efficiency, effectiveness, and error reduction. Modern Interaction Design has shifted to include **user experience (UX)**, specifically focusing on how interactive systems make people *feel*.

Emotional Interaction

Emotional interaction is concerned with how we feel and react when interacting with technology. It involves designing systems that elicit specific emotional responses (e.g., trust, happiness, motivation) and avoiding negative ones (e.g., frustration, anxiety).

1.1 Emotions and Behavior

There is a complex relationship between emotional state and cognitive processing. Research (including Baumeister et al., 2007) suggests:

- **Negative Emotions (Fear/Anger):** Cause people to focus narrowly. The body tenses; tolerance for errors decreases.
- **Positive Emotions (Happiness/Joy):** Cause people to relax. Users become more creative, better at problem-solving, and more likely to overlook minor interface glitches.

Real-World Example: The ATM Machine

If an ATM user is anxious (fear of losing money) or angry (in a rush), a confusing interface will cause immediate panic or rage. If the user is happy, they might laugh off a slow loading screen. Designers must account for the user's likely emotional state during critical tasks.

2 Three Levels of Design (Don Norman)

Donald Norman (author of *Emotional Design*) creates a framework connecting emotions to design through three cognitive levels. This is based on the Ortony et al. (2005) model.

2.1 1. Visceral Design

This is the pre-conscious level. It deals with the initial impact of a product's appearance.

- **Focus:** Look, feel, and sound.
- **Goal:** Make the product aesthetically pleasing to trigger a positive gut reaction.
- **Key Factor:** "Attractive things work better" (psychological phenomenon where aesthetics mask minor usability issues).

2.2 2. Behavioral Design

This deals with the use of the product and the experience of using it.

- **Focus:** Function, performance, and usability.
- **Goal:** Allow the user to feel in control and accomplish tasks effectively.
- **Key Factor:** Traditional Usability (Efficiency, Learnability).

2.3 3. Reflective Design

This is the highest cognitive level, involving conscious thought, memory, and self-image.

- **Focus:** Meaning, personal value, and cultural message.
- **Goal:** Create a product that says something about the user (status symbol) or evokes memories.
- **Key Factor:** Long-term emotional attachment.

Real-World Example: The Swatch Watch vs. Smartwatch

- **Visceral:** The bright colors and transparent case of a Swatch watch catch the eye.
- **Behavioral:** It tells time accurately and fits the wrist (usability).
- **Reflective:** Wearing it might signal a fun personality or retro fashion sense (meaning).

3 Expressive Interfaces

Interfaces use feedback to convey emotional states. This includes icons, sounds, animations, and color palettes.

3.1 Evolution of Expressive Feedback

- **1980s:** The "Happy Mac" icon indicated a successful boot, while a "Sad Mac" indicated a crash. This anthropomorphized the machine.
- **Modern:** Interfaces use abstract but aesthetic indicators, such as the spinning beachball (MacOS) or loading skeletons.

3.2 Designing Error Messages

Poorly designed error messages cause frustration. Ben Shneiderman provides classic guidelines to reduce negative emotional impact:

1. Avoid negative/condemning terms (FATAL, ILLEGAL, INVALID).
2. Avoid UPPERCASE (it feels like shouting).

3. Use precise, constructive language rather than vague codes (e.g., "Error 404").
4. Provide context-sensitive help.

Real-World Example: Lego's 404 Page

Instead of a dry "Page Not Found" text, the Lego website displays a scared Lego figure. This utilizes humor to diffuse the user's frustration, turning a negative experience into a neutral or positive one.

4 Affective Computing Emotional AI

Affective Computing (coined by Rosalind Picard, 1998) is the study and development of systems and devices that can recognize, interpret, process, and simulate human affects (emotions).

4.1 Techniques for Emotion Detection

- **Facial Coding:** Using cameras and AI (e.g., Affdex software) to map facial muscles to emotions (smiling = joy, brow furrow = anger).
- **Voice Analysis:** Analyzing pitch, tone, and volume to detect stress or anger (often used in call centers).
- **Biometrics:** Galvanic Skin Response (GSR) sensors on fingers/palms measure sweat/arousal; heart rate monitors measure stress.
- **Motion Capture:** Analyzing body language and gestures (e.g., slouching = boredom/sadness).

4.2 Plutchik's Wheel of Emotions

Robert Plutchik classified emotions into 8 basic categories arranged in a wheel:

1. Joy
2. Trust
3. Fear
4. Surprise
5. Sadness
6. Disgust
7. Anger
8. Anticipation

Designers use this to create color palettes or "mood boards" that align with specific emotional intensities.

4.3 Applications of Emotional AI

- **Marketing:** Analyzing reaction to ads (if a user looks disgusted, stop showing the ad).
- **Automotive:** Detecting road rage or fatigue in drivers and suggesting a break.
- **Health:** Detecting signs of depression through speech or typing patterns.

5 Behavioral Change and Persuasive Tech

Technology can be designed to nudge users toward "better" behaviors (Fogg, 2003).

5.1 Nudging

Nudging involves subtle design choices that alter behavior without forbidding options.

- **Pop-ups/Reminders:** "Time to stand up!" (Apple Watch).
- **Default Settings:** Opt-out organ donation vs. Opt-in.
- **Social Proof:** Showing how a user compares to peers.

5.2 Examples of Behavioral Technology

1. **Nintendo's Pocket Pikachu:** A pedometer game. If the child walks/jumps, the virtual pet is happy. If the child is sedentary, the pet becomes angry. This utilizes *emotional attachment* to drive physical activity.
2. **The Tidy Street Project:** A sustainable HCI project where street electricity usage was visualized on the road surface using chalk. This public display utilized *social pressure* and normative influence to reduce energy consumption by 15%.
3. **Mood Trackers (Daylio/Moodnotes):** Apps that allow users to log feelings. Reflection helps users identify triggers for bad moods, improving mental well-being.

6 Anthropomorphism

Anthropomorphism is the attribution of human qualities to inanimate objects. In HCI, this means designing computers, robots, or interfaces that appear to have personality.

6.1 Pros and Cons

- **Pros:** Makes interaction more enjoyable; reduces anxiety; helps users trust the system (e.g., educational software praising a student).
- **Cons:** Can be deceptive; can create false expectations of intelligence; can be annoying or "creepy" (The Uncanny Valley).

6.2 The Case of "Clippy" vs. "Anna"

- **Microsoft's Clippy:** Failed because it was intrusive. It interrupted work with obvious suggestions ("It looks like you're writing a letter"), causing frustration.
- **IKEA's Anna:** A virtual agent that used facial expressions (blinking, smiling). It was generally better received because it was less intrusive, though still limited.

6.3 Zoomorphism: Virtual Pets and Robots

Sometimes designing technology as an animal is more effective than as a human, as users have lower expectations of intelligence for animals.

- **Aibo (Sony):** A robot dog. Owners developed deep emotional bonds, treating it like a living pet.
- **Paro:** A therapeutic robot seal used in care homes. It reduces stress and encourages social interaction among patients with dementia.

Real-World Example: Ethical Dilemma: Robots in Care

Is it ethical for elderly people to form emotional attachments to robots like Paro or Stevie?

- **Argument For:** It provides comfort, reduces loneliness, and motivates patients when human staff are unavailable.
- **Argument Against:** It is "deceptive" and potentially demeaning to substitute human care with a machine that cannot truly return affection.

7 Summary

1. **Emotional Interaction** moves beyond efficiency to consider joy, frustration, and engagement.
2. **Design Models** (Visceral, Behavioral, Reflective) help designers target specific emotional responses.
3. **Expressive Interfaces** use aesthetics and polite feedback to manage user frustration.
4. **Affective Computing** uses sensors and AI to detect human emotion, raising both opportunities for personalization and privacy concerns.
5. **Persuasive Technology** uses emotional triggers (like virtual pets) and social pressure to change behavior.
6. **Anthropomorphism** is a powerful tool but requires balance—too much human-likeness can be creepy or annoying (Clippy), while zoomorphism (Paro) is often more accepted.